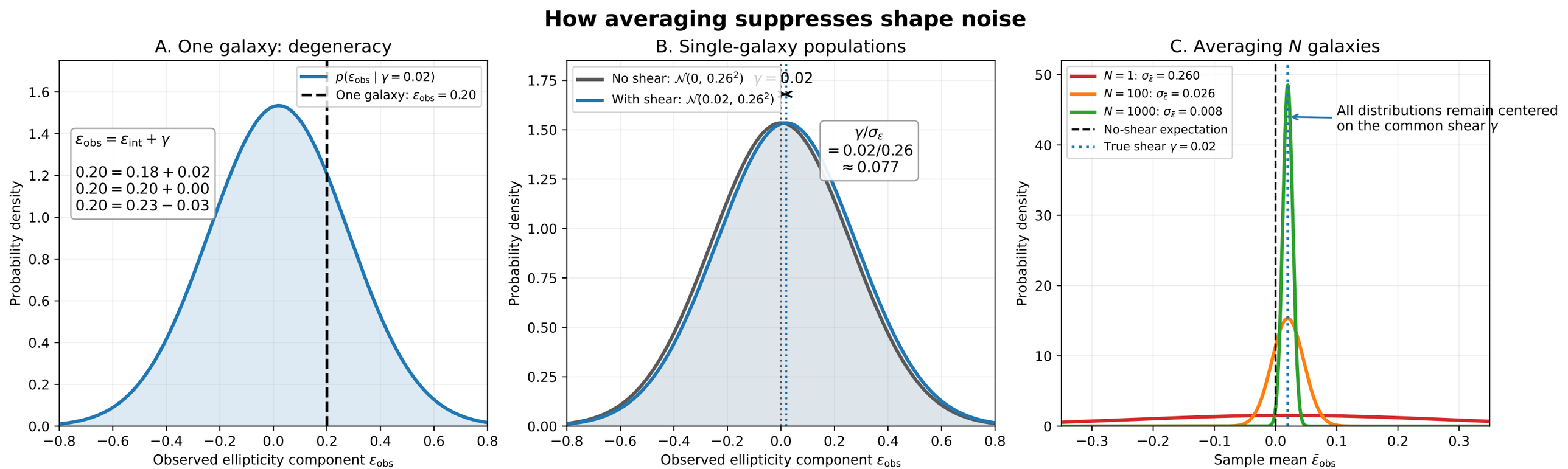

How averaging suppresses shape noise
A. One galaxy: degeneracy
$p(\varepsilon_{obs} \mid \gamma = 0.02)$
One galaxy: $\varepsilon_{obs} = 0.20$
$\varepsilon_{obs} = \varepsilon_{int} + \gamma$
$0.20 = 0.18 + 0.02$
$0.20 = 0.20 + 0.00$
$0.20 = 0.23 - 0.03$
Probability density
Observed ellipticity component $\varepsilon_{obs}$
B. Single-galaxy populations
No shear: $\mathcal{N}(0, 0.26^2)$
With shear: $\mathcal{N}(0.02, 0.26^2)$
$\gamma = 0.02$
$\gamma/\sigma_\varepsilon$
$= 0.02/0.26$
$\approx 0.077$
Probability density
Observed ellipticity component $\varepsilon_{obs}$
C. Averaging $N$ galaxies
$N = 1$: $\sigma_{\bar{\varepsilon}} = 0.260$
$N = 100$: $\sigma_{\bar{\varepsilon}} = 0.026$
$N = 1000$: $\sigma_{\bar{\varepsilon}} = 0.008$
No-shear expectation
True shear $\gamma = 0.02$
All distributions remain centered on the common shear $\gamma$
Probability density
Sample mean $\bar{\varepsilon}_{obs}$